Mariam Mohammed Mahmoud Abdelgawad

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OBJECTIVE

Computer Science & Engineering student at Egypt-Japan University of Science and Technology (E-JUST), with a strong academic record (CGPA 3.47) and hands-on experience in circuit design, digital logic, and IoT systems. Seeking opportunities to apply technical skills in Python, Verilog, MATLAB, and LTSpice within a challenging engineering environment, while continuing to grow through real-world projects and collaboration.

EDUCATION

Egypt-Japan University of Science and Technology (E-JUST) *Expected Graduation: 2028*
Faculty of Engineering | School of Electronics, Communications & Computer Engineering (ECCE)
B.Sc. in Computer Science & Engineering (CSE) | **Cumulative GPA:** 3.47 / 4.00

PROJECTS

Audio Amplifier using LM386 IC | *Electronics Lab*

- Designed and implemented an audio amplifier circuit using the LM386 IC.
- Applied analog circuit design principles and component integration techniques.

IoT-Based Solar Panel Performance Monitoring System | *IoT / Data Collection*

- Designed a real-time monitoring system for solar panel performance using IoT concepts and sensors.
- Implemented data collection and analysis workflows to evaluate energy output and efficiency.

Image Processing & Interpolation | *MATLAB*

- Implemented image scaling using Nearest Neighbour, Bilinear, and Bicubic interpolation methods.
- Compared interpolation techniques and analysed image quality using MATLAB.

Image Processing & Computer Vision | *Python / OpenCV*

- Developed an image resizing application using Python, OpenCV, and Matplotlib.
- Compared image quality across multiple interpolation methods.

Quiz Engine Application | *Python*

- Built an interactive quiz application with scoring, input validation, and modular programming concepts.

Engineering Your Future – Research Project | *Key Skills Course*

- Conducted a survey-based research project on factors affecting students' engineering major selection.
- Analysed survey results and presented recommendations for career guidance improvement.

TECHNICAL SKILLS

Programming & HDL: Python (problem solving & algorithms), Verilog HDL (digital logic design)
Simulation & Design Tools: MATLAB (signal processing & simulation), LTSpice (circuit simulation), SolidWorks (3D design)

RELEVANT COURSEWORK

• Digital Logic Design • Signals & Systems • Introduction to Electronics • Measurements & Instrumentation
• Computer Programming • Electrical Engineering Fundamentals • Mathematics for Engineering

KEY COMPETENCIES

• Teamwork & Collaboration • Problem Solving & Analytical Thinking • Time Management & Adaptability
• Fast Learner • Ability to Work Under Pressure

LANGUAGES

Arabic: Native | **English:** B2 – Upper Intermediate